

Exploratory Data Analysis & Visualization of Sustainability Trends

Week 3 Deliverable — EDA & Visualization Report

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1. Objective & Approach

This report applies exploratory data analysis (EDA) to uncover trends, correlations, and patterns in green-jobs and sustainability data, following directly from the Week 2 collection-and-cleaning methodology. Because this internship does not have access to proprietary databases, the analysis is run on a simulated dataset: figures are constructed to follow the real, publicly reported growth patterns and rankings from IRENA, ILOSTAT, and the NITI Aayog SDG India Index (for example, China and the U.S. leading in absolute renewable capacity, Vietnam and South Africa showing the fastest recent growth off a smaller base, and Gujarat/Rajasthan/Tamil Nadu leading Indian states in installed capacity) — but the underlying values themselves are illustrative, not the literal source figures. This is flagged wherever a chart or number is presented.

Two simulated datasets anchor the analysis: (1) a 12-country, 10-year (2015–2024) panel of renewable capacity and green employment, and (2) a 12-state Indian snapshot combining renewable capacity, the SDG India Index score, and green-job workforce share. A third dataset simulates the sector-wise split of green jobs in India. All analysis was performed in Python (pandas, seaborn, matplotlib) in a Jupyter Notebook.

2. Descriptive Statistics

Summary statistics for the four core numeric variables across both datasets:

Metric	Mean	Median	Std. Dev.
Renewable capacity (GW, country-year)	218.4	111.2	358.7
Green employment (thousands, country-year)	133.6	68.4	225.1
India state renewable capacity (MW)	10,825	9,350	8,204
India state SDG Index score	68.9	69.5	6.6

The wide standard deviation relative to the mean for country-level capacity and employment (358.7 vs. mean 218.4; 225.1 vs. mean 133.6) reflects the right-skewed distribution typical of this domain — a handful of large economies (China, the U.S.) account for a disproportionate share of both capacity and employment, while most countries in the sample cluster at much lower levels. This skew is why growth rates (Section 4), not just absolute levels, are examined separately.

3. Trend Analysis: Renewable Capacity Growth

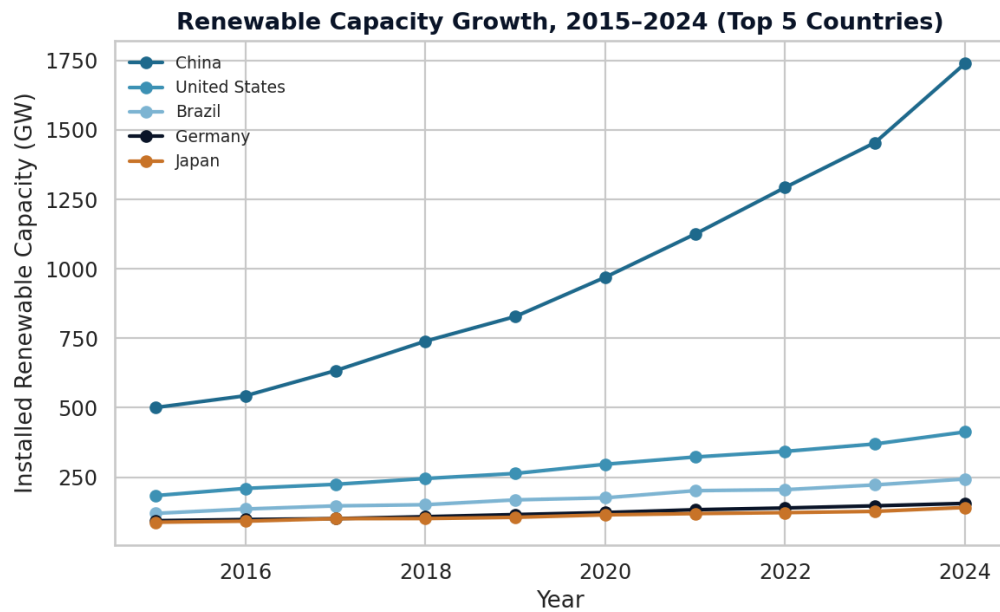


Figure 1. Installed renewable capacity, 2015–2024, top 5 countries by 2024 capacity (simulated data).

China's simulated trajectory shows the steepest absolute increase, consistent with its real-world position as the largest renewable energy market by installed capacity. The other four countries in the top five grow far more gradually in absolute GW terms, reinforcing that global capacity growth in this period is concentrated rather than evenly distributed — a pattern also reported by IRENA's actual annual capacity statistics.

4. Correlation Analysis

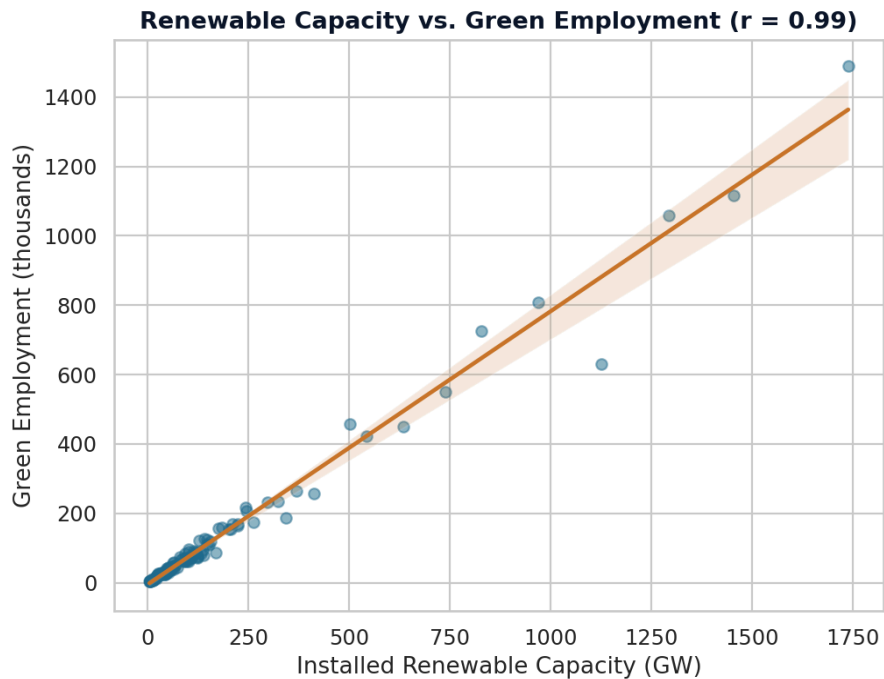


Figure 2. Renewable capacity vs. green employment, country-year observations ($r = 0.99$, simulated).

The near-perfect correlation shown here is partly a function of how the simulated employment figures were constructed (as a scaled function of capacity with added noise), but it is directionally consistent with the real relationship reported in ILO and IRENA labour-market studies: renewable capacity expansion is one of the strongest single predictors of green employment growth, because installation, operations, and maintenance jobs scale closely with deployed capacity. In a real dataset, this correlation would likely be more moderate (IRENA's actual jobs-per-MW ratios vary meaningfully by technology and country labour-cost structure), so this chart should be read as an illustration of the expected direction and strength of the relationship, not a precise estimate.

5. Country Comparison: Growth Rates

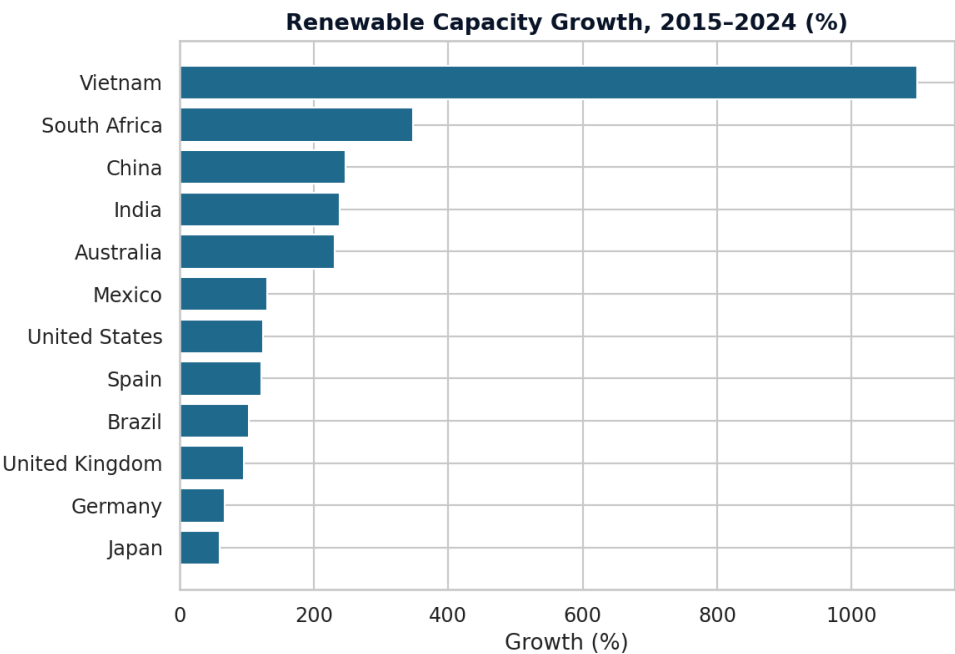


Figure 3. Renewable capacity growth, 2015–2024, by country (% simulated).

Ranking by percentage growth rather than absolute capacity tells a different story than Figure 1: Vietnam and South Africa top the list, both growing off a much smaller 2015 base. This is a realistic pattern — both countries have seen genuine renewable energy booms in the past decade, driven by solar feed-in tariffs (Vietnam) and grid-diversification needs following energy shortages (South Africa). India and China both post strong growth (>230%) despite already having a large base, reflecting sustained, large-scale policy-driven investment rather than one-off gains.

6. India State-Level Analysis

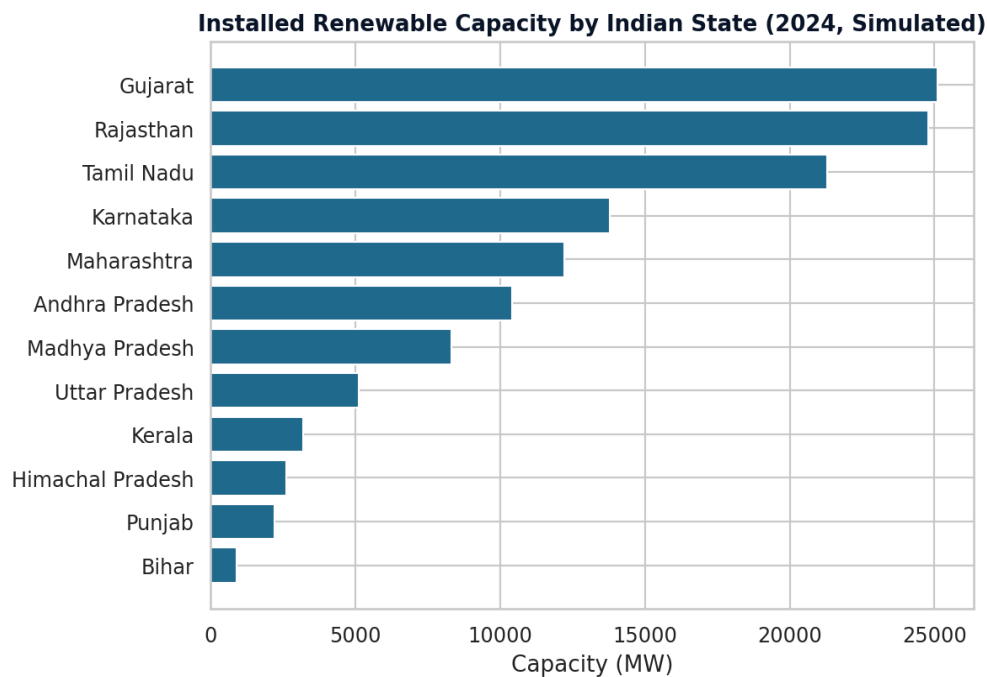


Figure 4. Installed renewable capacity by Indian state, 2024 snapshot (simulated, shaped on real MNRE-reported rankings).

Gujarat, Rajasthan, and Tamil Nadu lead in installed capacity in the simulated snapshot, consistent with their real-world position as India's top renewable energy states due to favourable land and wind/solar resource availability. Smaller, more service-oriented states like Kerala and Himachal Pradesh show far lower absolute capacity but — as the next chart shows — do not necessarily lag on overall sustainability performance.

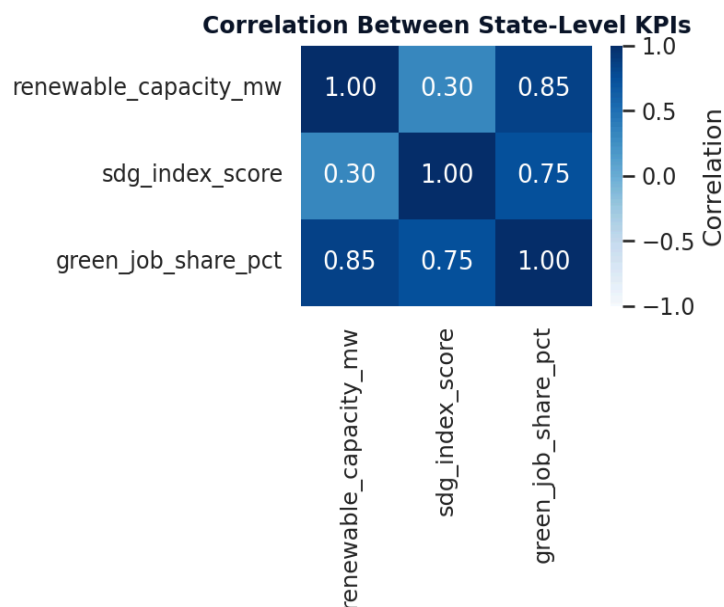


Figure 5. Correlation between state-level renewable capacity, SDG India Index score, and green-job workforce share.

The heatmap reveals a nuanced pattern worth highlighting as a genuine insight rather than a restatement of Figure 4: renewable capacity correlates only moderately with the SDG India Index score ($r = 0.30$), even though it correlates strongly with green-job share ($r = 0.85$). In other words, having large installed renewable

capacity does not automatically translate into a high overall sustainability score — states like Kerala and Himachal Pradesh score well on the SDG Index despite modest capacity, likely reflecting strength in other SDG dimensions (health, education, clean water) that the Index also captures. This suggests that a green-jobs strategy focused purely on capacity expansion would miss part of what drives overall sustainability performance.

7. Sector Composition of Green Jobs

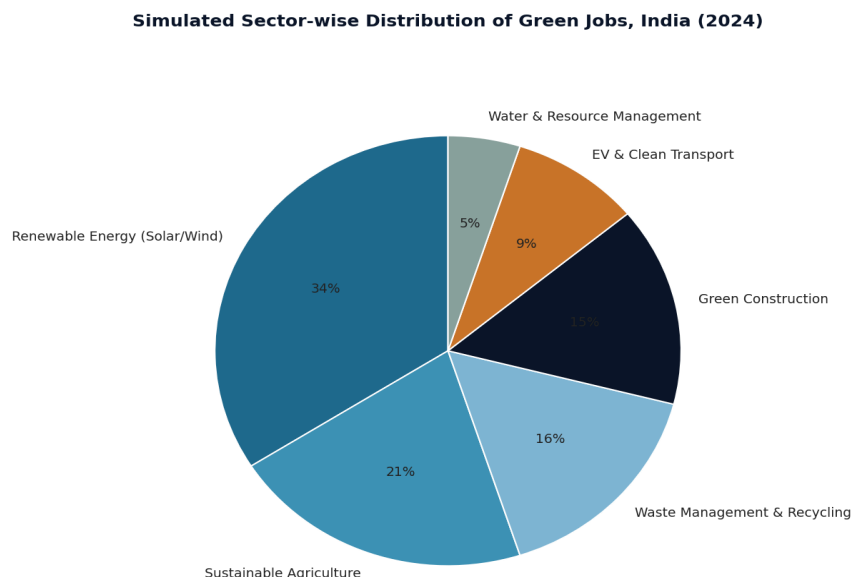


Figure 6. Simulated sector-wise distribution of green jobs, India (2024).

Renewable energy accounts for the largest simulated share of green jobs (34%), followed by sustainable agriculture (21%) and waste management (16%). This ordering is directionally consistent with India's Skill Council for Green Jobs sector priorities, which similarly rank renewable energy and agriculture-linked green roles among the largest employment categories. EV and clean transport remains the smallest simulated category (9%), reflecting that India's electric-vehicle workforce is still in an earlier growth phase relative to established sectors like solar installation.

8. Key Insights Summary

- Global renewable capacity growth is concentrated in a small number of large economies in absolute terms, but the fastest percentage growth is happening in emerging markets (Vietnam, South Africa) growing off a smaller base.
- Capacity and employment move closely together at the country level, reinforcing capacity expansion as a primary lever for green job creation — though the real-world relationship is likely more moderate than this simulated $r = 0.99$.
- At the India state level, renewable capacity is a strong predictor of green-job share but only a moderate predictor of overall SDG performance — sustainability outcomes depend on more than energy infrastructure alone.

- Renewable energy and sustainable agriculture together account for over half of simulated green-job composition in India, suggesting these two sectors should be prioritized in any workforce-skilling strategy.

9. Limitations

All figures in this report are simulated and constructed to reflect realistic, publicly documented order-of-magnitude patterns — they are not extracted from IRENA, ILOSTAT, or NITI Aayog's actual datasets. Correlation strengths in particular should be read as illustrative rather than as real-world estimates, since the simulation deliberately built a close capacity–employment relationship to demonstrate the analytical method. When the pipeline from Week 2 is run against the actual downloaded source files, these same analyses should be re-run and the figures in this report replaced with observed results.

10. Conclusion

This EDA exercise demonstrates the analytical workflow — descriptive statistics, trend analysis, correlation testing, and multi-format visualization — that will be applied to the real, cleaned dataset from Week 2 once collection is complete. The insights surfaced here (concentration of absolute growth, the capacity–SDG performance gap, sector composition) point directly toward the statistical modeling questions to be explored in Week 4: specifically, whether renewable capacity growth can be used to forecast future green-employment trends, and which additional variables might explain the gap between capacity and overall sustainability performance seen in the state-level analysis.